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Bayesian Phylolinguistic Inference under Cognate Dependence

A Study of Model Misspecification

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Background

Goal: Reconstruct language tree T^* from data $\mathbf{D}$ using Bayesian phylogenetic analysis

Language	'to laugh'		'wing'	
	Word	Class	Word	Class
Fijian	dredre	tl-A	taba-na	w-A
Marquesan	kata	tl-B	peheu	w-B
Hawaiian	'aka	tl-B	'heu	w-B
Maori	kata	tl-B	parirau	w-C
Tahitian	'ata	tl-B	pererau	w-C

Language	'to laugh'		'wing'		
	A	B	A	B	C
Fijian	1	0	1	0	0
Marquesan	0	1	0	1	0
Hawaiian	0	1	0	1	0
Maori	0	1	0	0	1
Tahitian	0	1	0	0	1

Table: Binary encoding of cognacy classes from five Oceanic languages. Adapted from Hoffmann et al. (2021).

Problematic Assumption: Conditional Site Independence

Research Questions

Question 1
How does unknown site dependence within data hurt tree reconstruction?

Question 2
Does adding dependent sites improve tree reconstruction?

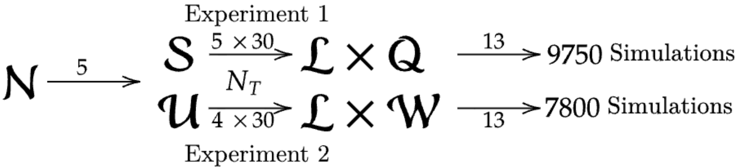
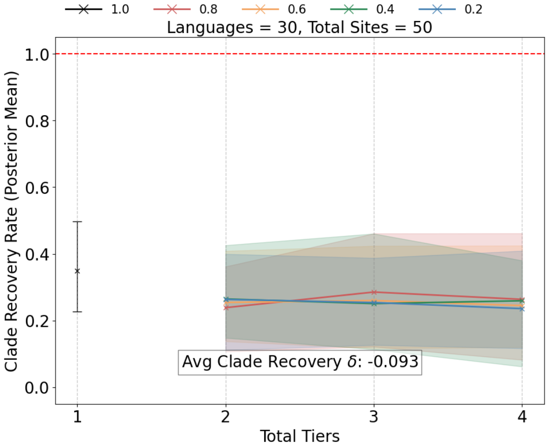


Figure: Factorial Design of experiments.

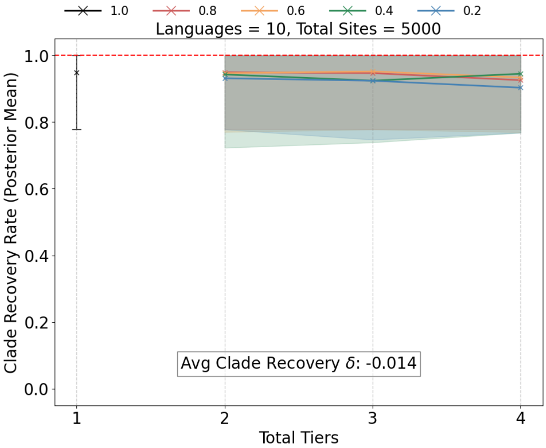
Results from Experiments

Experiment 1 (Unknown Dependency Structure)

Small Dataset



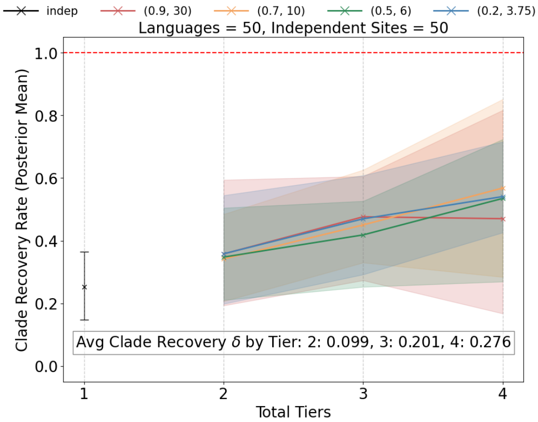
Large Dataset



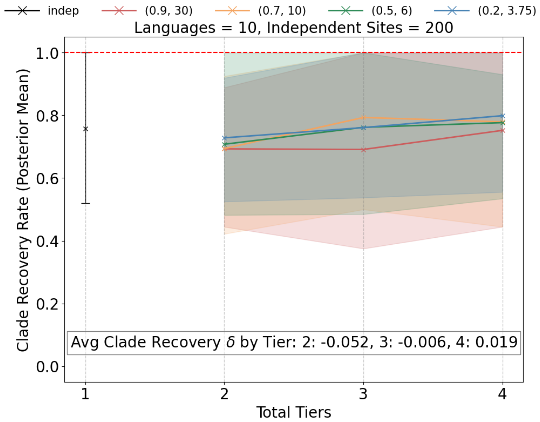
Results from Experiments

Experiment 2 (Additional Dependent Sites)

Small Dataset



Large Dataset

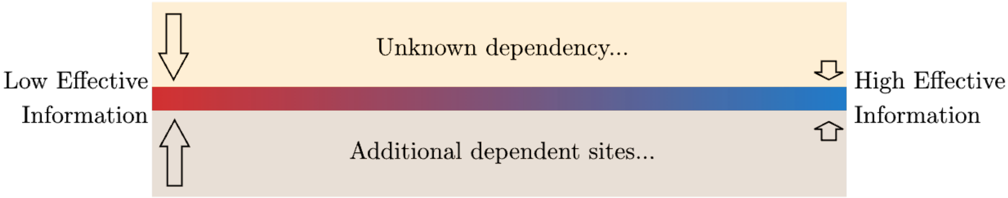


Key Result

Contradiction? Rather...

Effective Information

All dependent sites cause model misspecification, but...



Conclusion

Effective Information vs Model Misspecification

Experiment 1: Effects of unknown dependencies

Experiment 2: Decision-making on retaining dependent sites

References

K. Hoffmann, R. Bouckaert, S. J. Greenhill, and D. Kühnert. Bayesian phylogenetic analysis of linguistic data using BEAST. Journal of Language Evolution, 6(2):119–135, Sept. 2021. _eprint: https://academic.oup.com/jole/article-pdf/6/2/119/41325380/lzab005.pdf.

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Thank you!